ASSIGNMENT - 8.2

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Batch:29

TaskDescription
Task1–Test-DrivenDevelopmentforEven/OddNumberValidator
• UseAItoolstofirstgeneratetestcasesforafunctionis_even(n) and then implement the function so that it satisfies all generated tests.
Requirements:
• Input must be an integer
• Handlezero,negativenumbers,andlargeintegers
Example Test Scenarios:
is_even(2) → True
is_even(7) → False
is_even(0) → True
is_even(-4) → True
is_even(9) → False
Expected Output -1
• Acorrectlyimplementedis_even()functionthatpassesallAI-generated test cases

task1.py > ...

```python
# ------------------------------
# Test-Driven Development (TDD)
# Even/Odd Number Validator
# ------------------------------


# Step 1: Define test cases
def test_is_even():
    # Positive numbers
    assert is_even(2) == True
    assert is_even(7) == False

    # Zero
    assert is_even(0) == True

    # Negative numbers
    assert is_even(-4) == True
    assert is_even(-9) == False

    # Large integers
    assert is_even(10**15) == True
    assert is_even(10**15 + 1) == False

    # Invalid input (should raise TypeError)
    try:
        is_even("10")
    except TypeError:
        pass
    else:
        assert False, "TypeError not raised for string input"


# Step 2: Implement function
def is_even(n):
    if not isinstance(n, int):
        raise TypeError("Input must be an integer")
    return n % 2 == 0


# Step 3: Run tests
if __name__ == "__main__":
    test_is_even()
    print("All test cases passed successfully!")

```

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Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                    Python

/usr/bin/python3 "/Users/arepallyramcharan/ai assistent.py/assignment - 8.2.py/task1.py"
arepallyramcharan@Arepallys-MacBook-Air assignment - 8.2.py % /usr/bin/python3 "/Users/arepallyramcharan/ai assi
stent.py/assignment - 8.2.py/task1.py"
All test cases passed successfully!
arepallyramcharan@Arepallys-MacBook-Air assignment - 8.2.py %
```

Observation:

TheTest-DrivenDevelopment(TDD)approachensuresthattestcasesare written before implementing the function, improving code reliability.

Writingassert-basedtest caseshelped verifythe correctnessof the `is_even(n)` function for positive, negative, zero, and large integer inputs.

Inputvalidationusing`isinstance()` ensuresthatonlyintegervaluesare accepted, and a `TypeError`is raised for invalid inputs.

Themodulusoperator(`n%2==0`)correctlydeterminesevennumbers, including negative integers.

Runningalltestcasessuccessfullyconfirmsthatthefunctionsatisfiesfunctional requirements and handles edge cases properly.

TaskDescription

Task2–Test-DrivenDevelopmentforStringCaseConverter

- **Ask AI to generate test cases for two functions:**
- **to_uppercase(text)**
- **to_lowercase(text)**

Requirements:
• Handle empty strings
• Handle mixed-case input
• HandleinvalidinputssuchasnumbersorNone
Example Test Scenarios:
to_uppercase("aicoding")→"AICODING"
to_lowercase("TEST") → "test"
to_uppercase("")→""
to_lowercase(None)→Errororsafehandling Expected Output -2
• TwostringconversionfunctionsthatpassallAI-generatedtest cases with safe input handling.

task2.py > ...

```python
def test_string_case_functions():
        else:
            assert False, "TypeError not raised for integer input"

    try:
        to_lowercase(None)
    except TypeError:
        pass
    else:
        assert False, "TypeError not raised for None input"


# Step 2: Implement Functions
def to_uppercase(text):
    if not isinstance(text, str):
        raise TypeError("Input must be a string")
    return text.upper()


def to_lowercase(text):
    if not isinstance(text, str):
        raise TypeError("Input must be a string")
    return text.lower()


# Step 3: Run Tests
if __name__ == "__main__":
    test_string_case_functions()
    print("All test cases passed successfully!")
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                    Python

/usr/bin/python3 "/Users/arepallyramcharan/ai assistent.py/assignment - 8.2.py/task2.py"
arepallyramcharan@Arepallys-MacBook-Air assignment - 8.2.py % /usr/bin/python3 "/Users/arepallyramcharan/ai assi
stent.py/assignment - 8.2.py/task2.py"
All test cases passed successfully!
arepallyramcharan@Arepallys-MacBook-Air assignment - 8.2.py %
```

Observation:

1. TheTest-DrivenDevelopment(TDD)approachensuresthattestcasesare written before implementing the string conversion functions.

2. The `.upper()` and `.lower()` built-instringmethodscorrectlyconverttextwhile preserving non-alphabet characters.

3. Edge cases such as empty strings are handled safely without errors.

4. Inputvalidationusing`isinstance(text,str)` ensuresthatinvalidinputslike integers or `None`raise a `TypeError`.

5. Successfulexecutionofalltestcasesconfirmsthatbothfunctionsmeet functional and safety requirements.

TaskDescription
Task3–Test-DrivenDevelopmentforListSumCalculator
• UseAItogeneratetestcasesforafunctionsum_list(numbers) that calculates the sum of list elements.
Requirements:
• Handle empty lists

• Handle negative numbers
• Ignoreorsafelyhandlenon-numericvalues
Example Test Scenarios:
sum_list([1, 2, 3]) → 6
sum_list([]) → 0
sum_list([-1, 5, -4]) → 0
sum_list([2, "a", 3]) → 5
Expected Output 3
• ArobustlistsumfunctionvalidatedusingAI-generatedtest cases.

task3.py > ...

```python
# ----------------------------------------
# Test-Driven Development (TDD)
# List Sum Calculator
# ----------------------------------------


# Step 1: Define Test Cases
def test_sum_list():

    # Normal case
    assert sum_list([1, 2, 3]) == 6

    # Empty list
    assert sum_list([]) == 0

    # Negative numbers
    assert sum_list([-1, 5, -4]) == 0

    # Mixed numeric and non-numeric values
    assert sum_list([2, "a", 3]) == 5

    # Large numbers
    assert sum_list([10**10, 10**10]) == 2 * 10**10

    # List with only non-numeric values
    assert sum_list(["a", None, "b"]) == 0


# Step 2: Implement Function
def sum_list(numbers):

    if not isinstance(numbers, list):
        raise TypeError("Input must be a list")
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS                    Python

/usr/bin/python3 "/Users/arepallyramcharan/ai assistent.py/assignment - 8.2.py/task3.py"
arepallyramcharan@Arepallys-MacBook-Air assignment - 8.2.py % /usr/bin/python3 "/Users/arepallyramcharan/ai assi
stent.py/assignment - 8.2.py/task3.py"
All test cases passed successfully!
arepallyramcharan@Arepallys-MacBook-Air assignment - 8.2.py %
```

Observation:

1. Test-DrivenDevelopment(TDD)ensuresthattestcasesarewrittenbefore implementing the `sum_list()` function.

2. Thefunctioncorrectlyhandlesemptylistsbyreturning `0`,preventingruntime errors.

3. Negative numbers are properly included in the summation logic.

4. Non-numericvaluessuchasstringsand`None`aresafelyignored,improving robustness.

5. Input validation ensures that only list inputs are accepted, raising a `TypeError` for invalid input types.

TaskDescription
Task4–TestCasesforStudentResultClass
• GeneratetestcasesforaStudentResultclasswiththefollowing
methods:
• add_marks(mark)
• calculate_average()
• get_result()
Requirements:
• Marks must be between 0 and 100
• Average≥40→Pass,otherwiseFail

ExampleTestScenarios:
Marks:[60,70,80]→Average:70→Result:Pass
Marks:[30,35,40]→Average:35→Result:Fail Marks: [-10] → Error
Expected Output -4
• AfullyfunctionalStudentResultclassthatpassesallAI-generated test

task4.py > ...

```python
# ----------------------------------------
# Test-Driven Development (TDD)
# Student Result Class
# ----------------------------------------


# Step 1: Define Test Cases
def test_student_result():

    # Test Case 1: Pass scenario
    student1 = StudentResult()
    student1.add_marks(60)
    student1.add_marks(70)
    student1.add_marks(80)

    assert student1.calculate_average() == 70
    assert student1.get_result() == "Pass"

    # Test Case 2: Fail scenario
    student2 = StudentResult()
    student2.add_marks(30)
    student2.add_marks(35)
    student2.add_marks(40)

    assert student2.calculate_average() == 35
    assert student2.get_result() == "Fail"

    # Test Case 3: Invalid negative marks
    student3 = StudentResult()
    try:
        student3.add_marks(-10)
    except ValueError:
        pass
    else:
        assert False, "ValueError not raised for negative marks"


# Step 2: Implement StudentResult class
class StudentResult:
    def __init__(self):
        self._marks = []

    def add_marks(self, mark):
        if not isinstance(mark, (int, float)):
            raise TypeError("Mark must be a number")
```

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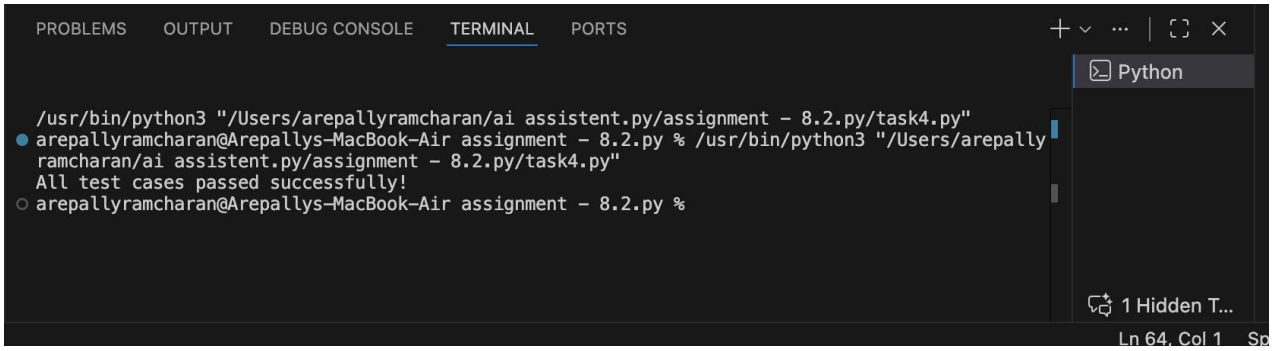
Output:



Observation:

1. TheTest-DrivenDevelopment(TDD)approachensuresallclassmethodsare validated before final implementation.

2. The`add_marks()`methodenforcesstrictvalidation,allowingonlymarks between 0 and 100.

3. The`calculate_average()`methodsafelyhandlesemptymarklistsby returning 0.

4. The`get_result()`methodcorrectlydeterminesPassorFailbasedonthe average threshold of 40.

5. Exceptionhandlingimprovesrobustnessbypreventinginvaliddatafrombeing stored.

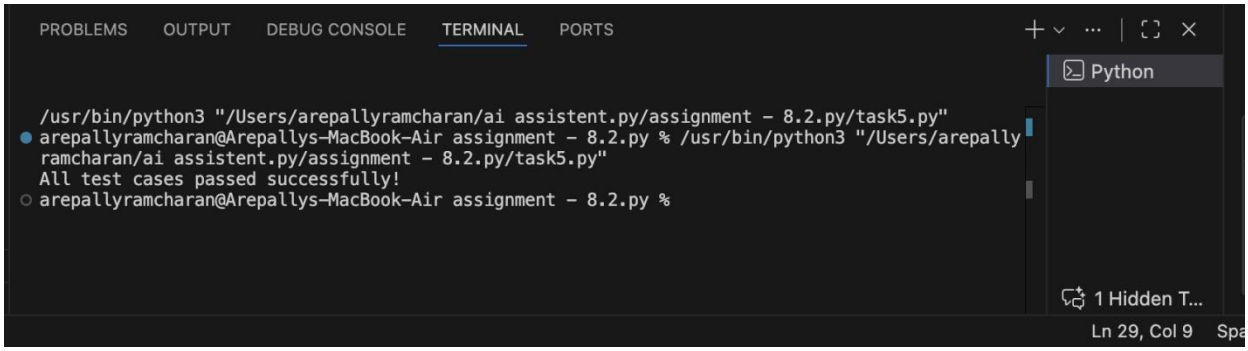
TaskDescription
Task5–Test-DrivenDevelopmentforUsernameValidator Requirements:
• Minimum length: 5 characters
• No spaces allowed
• Only alphanumeric characters

Example Test Scenarios:
is_valid_username("user01") → True
is_valid_username("ai") → False
is_valid_username("username")→False
is_valid_username("user@123")→False Expected
Output 5
A username validation function that passes all AI-generated test cases.

task5.py > ...

```python
# ----------------------------------------
# Test-Driven Development (TDD)
# Username Validator
# ----------------------------------------


# Step 1: Define Test Cases
def test_is_valid_username():

    # Valid username
    assert is_valid_username("user01") == True

    # Minimum length check
    assert is_valid_username("ai") == False
    assert is_valid_username("user1") == True  # exactly 5 characters

    # No spaces allowed
    assert is_valid_username("user name") == False

    # Only alphanumeric characters
    assert is_valid_username("user@123") == False
    assert is_valid_username("user_123") == False

    # Edge cases
    assert is_valid_username("") == False
    assert is_valid_username("12345") == True

    # Invalid input type
    try:
        is_valid_username(None)
    except TypeError:
        pass
    else:
```

Output:



Observation:

- Test-DrivenDevelopment(TDD)ensuresvalidationrulesareclearlydefined before implementing the username validator.
- Thefunctionenforcesaminimumlengthof5characterstomaintainbasic security standards.
- The `.isalnum()` method ensures only alphanumeric characters are allowed.
- Space detection prevents invalid usernames with whitespace.
- Typecheckingimprovesrobustnessbyraising`TypeError`fornon-stringinputs.